

IE University

School of Science and Technology

Cross-Frequency Aggregation in Mixed-Frequency Time Series

Forecasting US Consumer Energy Prices with Weekly Oil Data

Research Capstone

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AI Use Declaration

I acknowledge the use of AI-assisted tools during the preparation of this research capstone. These tools were used to support coding, debugging, structuring, editing, and clarification of written explanations. All research decisions, model choices, empirical interpretation, and final responsibility for the content of this thesis remain my own.

All code, results, tables, figures, and written conclusions were reviewed and validated by the author.

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Abstract

This thesis examines whether mixed-frequency models can improve forecasts of monthly US consumer energy prices by preserving the weekly timing of oil price movements. The main research question is whether models that use weekly oil-price information directly forecast monthly consumer energy prices better than models that first pre-aggregate weekly oil data into a monthly measure. The dependent variable is the monthly log-change in the US Consumer Energy CPI, while the main predictor is the weekly log-change in WTI crude oil prices.

The empirical analysis uses data from February 2000 to April 2026. Model development is first carried out through an expanding-window forecasting exercise over January 2015 to December 2022, and is then extended with a genuinely later external holdout from January 2023 to April 2026. The model comparison includes a monthly ARIMAX benchmark, parametric MIDAS, unrestricted MIDAS, a composite-link state-space model, LASSO-MIDAS, Kernel U-MIDAS, XGBoost, and an LSTM.

The results show that preserving weekly timing contains useful forecasting information. On the post-2022 external holdout, MIDAS with normalized beta weights is the best-performing model, reducing RMSE by 16.0% relative to ARIMAX. More broadly, the results show that mixed-frequency models remain competitive even when compared with more flexible machine-learning benchmarks. Robustness checks further suggest that the result is not simply driven by multicollinearity, dimensionality reduction, or overfitting. Across several specifications, the most informative oil-price signal is concentrated around prior-month weeks 2–3, pointing to an interpretable transmission window of roughly 5–7 weeks from crude oil prices to consumer energy prices.

1 Introduction

1.1 Context

Many economic time series are observed at different frequencies. Financial and energy market variables are often available daily or weekly, while macroeconomic outcomes such as inflation, GDP, or consumer prices are usually published monthly or quarterly. This creates a mixed-frequency problem: the researcher may have useful high-frequency information, but the target variable is observed only at a lower frequency. This problem is central in mixed-frequency forecasting, where methods such as MIDAS and state-space models aim to use high-frequency information without first collapsing it into a lower-frequency average (Aruoba et al., 2009; Ghysels et al., 2004, 2007; Mariano & Murasawa, 2003).

The starting point of this research capstone was the project brief *Exact Cross-Frequency Aggregation in Mixed-Frequency Time Series*. The original objective was to study low-frequency outcomes that depend on higher-frequency covariates and to develop a framework that avoids ad hoc pre-aggregation. Standard approaches such as ARIMAX usually aggregate the high-frequency variable before estimation, while MIDAS models preserve higher-frequency observations but often rely on restrictive lag-weight specifications. The proposed methodological direction was a composite-link state-space model, or CLM-SS, in which high-frequency observations are linked to the low-frequency target through an explicit aggregation matrix and autoregressive errors are handled in state-space form.

The original project brief was intentionally broad. It mentioned several possible empirical applications, including monthly inflation with daily or weekly oil prices, as well as electricity demand with weather and fuel-cost variables. In the first supervisor meeting, this broad direction was narrowed into two practical questions. The first was: *what are the natural competitors to MIDAS?* This mattered because MIDAS should not be evaluated in isolation. A credible capstone needed to compare MIDAS against standard monthly aggregation, unrestricted lag models, state-space alternatives, penalized regressions, and modern machine-learning benchmarks. The second question was more applied: *in a real coding and forecasting setting, is MIDAS competitive enough to be useful?* This shifted the project toward an empirical forecasting exercise rather than only a methodological discussion.

In the final capstone, I focus on the energy-price application because it provides a clear economic transmission mechanism and uses publicly available data. Crude oil prices are observed at a higher frequency than consumer energy prices, and changes in crude oil prices are expected to pass through to gasoline, heating oil, electricity, and other consumer energy components with some delay. This makes the setting appropriate for comparing ARIMAX, MIDAS, CLM-SS, and other forecasting alternatives. The link between oil prices and macroeconomic outcomes is well established in the literature, especially in work on oil shocks, inflation, and energy-price transmission (Baumeister & Hamilton, 2019; Hamilton, 1983; Kilian, 2009).

1.2 Data Selection and Motivation

The empirical application uses three public energy-related series. The target variable is the US Consumer Energy CPI from FRED, which measures the consumer-facing price of energy goods and services. The main high-frequency predictor is weekly WTI crude oil prices from FRED. A third series, the IMF Global Price of Energy Index, is used as a contextual energy-price indicator during the exploratory analysis, but it is not used as a forecasting predictor in the final models.

The first step was to evaluate whether these series were economically and empirically connected. In levels, Consumer Energy CPI and WTI crude oil prices move together over the sample

period, and the IMF energy index is also strongly related to both series. The exploratory analysis showed high correlations among the energy variables, with WTI and Consumer Energy CPI having a correlation of approximately 0.87 in levels. **Figure 1** visualizes this relationship. This does not prove a causal relationship, but it supports the idea that oil prices contain useful information for consumer energy prices. The forecasting models are estimated using log-differences rather than levels, because the level series are non-stationary while the transformed series are stationary.

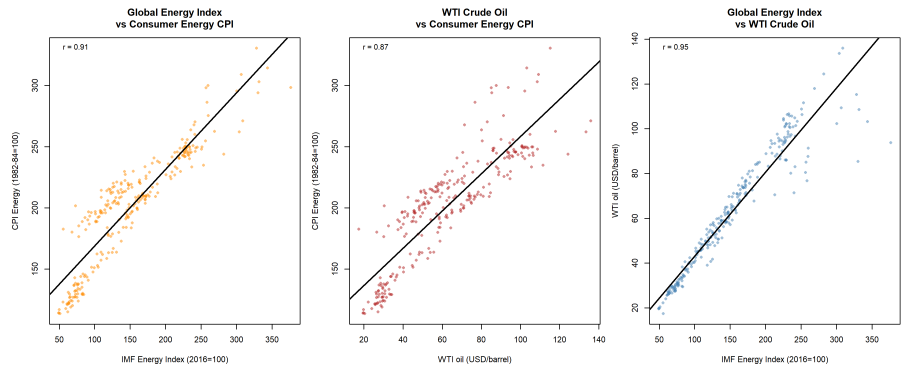


Figure 1: *Exploratory relationship between WTI crude oil prices, the IMF Global Price of Energy Index, and the US Consumer Energy CPI. The scatter plots show that the energy variables are strongly related in levels, supporting the use of oil prices as an informative predictor for consumer energy prices.*

Figure 2 shows the level relationship between the main energy variables over time. The figure motivates the empirical setting: oil prices and consumer energy prices are clearly related, but they are not identical and do not move one-for-one. This is exactly where a mixed-frequency model can be useful. Instead of asking only whether oil prices matter, the thesis asks whether the weekly timing of oil price changes matters for forecasting monthly consumer energy prices.

This choice of data is also appropriate for MIDAS. The dependent variable is monthly, while the predictor is weekly. In the empirical design, each month is represented by four weekly WTI observations. This creates a simple but meaningful mixed-frequency structure: the model can compare the role of different weeks instead of collapsing the entire month into one average oil-price change. If specific weekly lags are more informative than others, MIDAS and related models should be able to recover that timing pattern.



Figure 2: Monthly US Consumer Energy CPI and WTI crude oil prices over the sample period. The figure motivates the link between oil market movements and consumer energy prices, while also showing that the two series are not identical.

1.3 Problem Statement and Research Question

A standard monthly ARIMAX model handles the frequency mismatch by averaging weekly WTI changes into a single monthly oil-price variable. This is simple, but it may lose information. If consumer energy prices respond more strongly to oil price changes from specific weeks, then monthly averaging hides the timing of the transmission.

MIDAS models address this issue by allowing several weekly oil-price observations to enter the monthly forecasting equation with estimated lag weights (Ghysels et al., 2004, 2007). However, parametric MIDAS models impose a specific lag shape, such as normalized beta or exponential Almon weights. U-MIDAS relaxes this restriction by estimating separate weekly coefficients, but this can be unstable in small monthly samples (Feroni et al., 2015). The CLM-SS framework provides another perspective by estimating direct weekly link weights while allowing the remaining error to follow an autoregressive process, drawing on the state-space tradition in mixed-frequency modeling (Aruoba et al., 2009; Mariano & Murasawa, 2003).

The central research question of this thesis is:

Can mixed-frequency models that preserve weekly oil price timing improve forecasts of monthly US consumer energy prices relative to models that pre-aggregate weekly oil data?

This main question is divided into five sub-questions:

1. *What are the natural empirical competitors to MIDAS in this setting?*
2. *Does preserving weekly WTI information improve forecast accuracy compared with a monthly ARIMAX benchmark?*
3. *Which specification performs best: parametric MIDAS, U-MIDAS, CLM-SS, LASSO-MIDAS, Kernel U-MIDAS, XGBoost, or LSTM?*
4. *Do the models identify an interpretable oil-to-consumer-energy-price transmission window?*
5. *Does additional model complexity improve forecasting performance enough to justify the loss of interpretability?*

These subquestions guide the empirical structure of the thesis. The first question is addressed in the literature review and model comparison design, where MIDAS is positioned against econometric, state-space, penalized, and machine-learning alternatives such as XGBoost and LSTM (Chen & Guestrin, 2016; Fischer & Krauss, 2018). The second and third questions are answered through both the original model-development forecast exercise and a later external holdout. The fourth question is examined through the estimated MIDAS lag weights, CLM-SS link weights, LASSO coefficients, Kernel U-MIDAS weights, and XGBoost feature-importance results. The fifth question motivates the final performance-versus-interpretability comparison.

1.4 Objectives

The thesis has four main objectives. First, it identifies the relevant competitors to MIDAS in this empirical setting and compares them using a common forecasting design. Second, it evaluates whether preserving weekly timing improves forecast accuracy relative to monthly aggregation. Third, it studies whether different models identify a similar oil-to-consumer-price transmission window. Fourth, it assesses the trade-off between forecast accuracy, model complexity, and interpretability.

The empirical design now has three layers. The first is the original expanding-window model-development exercise from January 2015 to December 2022, with an initial training period from February 2000 to December 2014. The second is a genuinely later external holdout from January 2023 to April 2026, where the first forecast is estimated only with information available through December 2022 and the estimation window then expands over time. The third layer is a set of true future scenario forecasts beginning in July 2026, with short-horizon windows of 1, 3, and 6 months for the main thesis discussion, and a 12-month extension reserved for the appendix.

Forecast performance is evaluated mainly using RMSE, MAE, MASE, sMAPE, and directional accuracy. MAPE is reported with caution because the dependent variable is a monthly log-change and can be close to zero, which mechanically inflates percentage-based errors. The future forecasts are not evaluated against realized CPI values, since those observations are not yet available; instead, they are interpreted as scenario-conditioned projections based on alternative assumed oil-price paths.

1.5 Preview of Results

The results show that weekly timing matters, but in a more nuanced way than the initial version of the project suggested. In the original 2015–2022 model-development exercise, mixed-frequency models were already competitive with the monthly ARIMAX benchmark. However, the strongest validation comes from the later external holdout over January 2023 to April 2026. In that genuinely later period, MIDAS with normalized beta weights achieves the lowest RMSE, reducing forecast error by 16.0% relative to ARIMAX. This is an important result because it shows that the mixed-frequency advantage survives outside the original development window.

At the same time, the broader combined 2015–2026 robustness summary shows that machine-learning methods, especially XGBoost, are also highly competitive. This means the thesis does not support a simplistic conclusion that one model dominates in every setting. Instead, it shows that model rankings depend on the evaluation frame: XGBoost performs best in the full combined robustness summary, while MIDAS with normalized beta weights performs best in the cleaner post-2022 external holdout. This makes the comparison more informative rather than weaker.

As shown in **Figure 3**, mixed-frequency and flexible nonlinear models can both reduce forecast error relative to the monthly ARIMAX benchmark. A key contribution of the thesis is that it does not only compare models by forecast accuracy. It also studies what the models imply about the timing of oil-price transmission. Across several methods, including MIDAS, CLM-SS, LASSO-MIDAS, Kernel U-MIDAS, and XGBoost, the strongest signal appears around prior-month weeks 2–3. This points to an interpretable transmission window of roughly 5–7 weeks between movements in crude oil prices and changes in consumer energy prices. In this sense, the better-performing models are not simply more flexible; they also recover an economically meaningful lag structure.

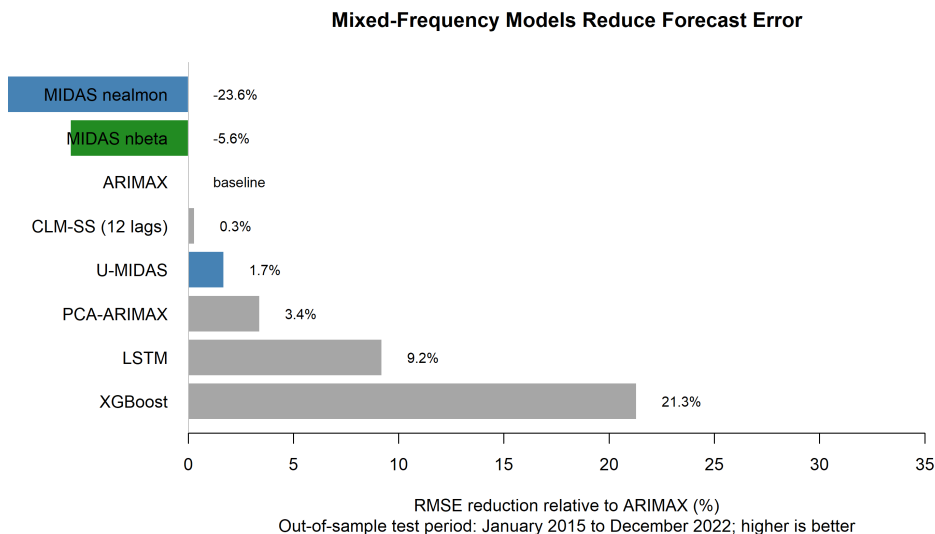


Figure 3: Out-of-sample RMSE reduction relative to the ARIMAX benchmark in the original model-comparison exercise. The figure illustrates that several mixed-frequency and machine-learning models improve on simple monthly aggregation, motivating the deeper evaluation carried out later in the thesis.

The rest of the thesis is organized as follows. Section 2 reviews the relevant literature and identifies the research gap. Section 3 describes the data, model specifications, and evaluation design. Section 4 presents the main empirical results. Section 5 discusses the interpretation and limitations. Section 6 concludes and suggests directions for future work. Supplementary forecast tables, lag-weight estimates, diagnostic figures, scenario projections, and code details are collected in the appendices.

2 Literature Review

This chapter reviews the main literature related to mixed-frequency forecasting, MIDAS models, state-space approaches, machine-learning benchmarks, and oil-price transmission to consumer prices. The goal is not only to summarize previous methods, but also to identify the gap that motivates this thesis: whether weekly oil-price information can improve forecasts of monthly consumer energy prices in a way that is both accurate and interpretable.

2.1 The Mixed-Frequency Forecasting Problem

A common issue in empirical macroeconomics and energy economics is that variables are observed at different frequencies. Consumer price indices are usually published monthly, while commodity prices such as crude oil are available daily or weekly. The simplest response is to aggregate the high-frequency variable to the lower frequency before estimation. For example, weekly WTI crude oil changes can be averaged into one monthly value and then used in a standard monthly regression or ARIMAX model.

This approach is convenient, but it can lose information. If oil price changes in some weeks matter more than others, then a simple monthly average hides the timing of the signal. This is especially relevant for consumer energy prices because the pass-through from crude oil prices to retail energy prices is unlikely to be immediate. Distribution costs, contracts, inventories, taxes, and retail pricing decisions can all create delays between movements in crude oil prices and the prices paid by consumers.

The mixed-frequency forecasting literature addresses this problem by allowing low-frequency outcomes to depend directly on high-frequency predictors. MIDAS regressions, introduced by Ghysels et al. (2004) and developed further by Ghysels et al. (2007), are one of the main approaches. Instead of pre-aggregating the high-frequency variable, MIDAS estimates a lag-weight function

that determines how much each high-frequency observation contributes to the low-frequency outcome.

2.2 Conceptual Comparison of Aggregation Approaches

The main difference between the methods reviewed in this chapter is how they treat the high-frequency predictor before it enters the forecasting equation. A pre-aggregated ARIMAX model first collapses weekly information into one monthly value:

$$\bar{x}_t = \frac{1}{m} \sum_{j=1}^m x_{t,j}, \quad (1)$$

where $x_{t,j}$ is the weekly WTI change in week j of month t , and $m = 4$ is the number of weekly observations used per month. The monthly forecasting equation can then be written as

$$y_t = \alpha + \phi y_{t-1} + \beta \bar{x}_t + \varepsilon_t. \quad (2)$$

This is simple, but it assumes that all weeks contribute equally. MIDAS models avoid this by estimating a weighted high-frequency lag structure:

$$y_t = \alpha + \sum_{k=0}^K w(k; \theta) x_{t-k/m} + \varepsilon_t, \quad (3)$$

where $w(k; \theta)$ is a parametric lag-weight function. This allows the model to learn whether some weekly lags matter more than others, but the result still depends on the chosen functional form (Ghysels et al., 2007).

U-MIDAS relaxes this by estimating each lag coefficient freely:

$$y_t = \alpha + \sum_{k=0}^K \beta_k x_{t-k/m} + \varepsilon_t. \quad (4)$$

This is more flexible, but it uses more parameters and may become unstable in small monthly samples (Foroni et al., 2015). The CLM-SS approach used in this thesis can be written as

$$y_t = \mu + Z_t w + u_t, \quad u_t = \rho u_{t-1} + \eta_t, \quad (5)$$

where Z_t contains the relevant weekly WTI observations, w is the vector of weekly link weights, and u_t is an autoregressive error component. This formulation preserves exact weekly timing, estimates flexible link weights, and allows the remaining error to have time-series structure.

The conceptual trade-off is already visible here. Simpler models gain transparency and stability by imposing aggregation in advance, while richer mixed-frequency models preserve timing information at the cost of additional modeling choices. This trade-off is central to the thesis, because the empirical contribution is not only to compare forecast accuracy, but also to evaluate whether better forecasts come from economically interpretable timing structure rather than from complexity alone.

2.3 Pre-Aggregation Benchmarks

ARIMAX models are the natural baseline in this thesis. They represent the standard monthly aggregation approach: weekly WTI changes are averaged into one monthly value and used as an exogenous regressor in a monthly time-series model. The strength of ARIMAX is that it is simple, transparent, and easy to estimate. Its weakness is that the aggregation rule is imposed before estima-

tion. ARIMA-style time-series modeling remains a standard benchmark in forecasting, originating from the classical Box-Jenkins framework (Box & Jenkins, 1970).

Bridge equations, discussed by Baffigi et al. (2004), are related. They also transform high-frequency indicators into lower-frequency predictors before estimation. Bridge models are useful in many macroeconomic nowcasting applications, but they still depend on a pre-specified aggregation rule. For this thesis, they do not solve the main problem, because the research question is specifically about whether the timing of weekly observations matters.

2.4 The MIDAS Regression Family

MIDAS models are the main econometric alternative to pre-aggregation. They allow high-frequency observations to enter a low-frequency forecasting equation through a lag-weight function. This makes them especially useful when the researcher believes that the timing of the high-frequency information is important (Ghysels et al., 2004, 2007).

Two common parametric MIDAS weight functions are used in this thesis. The exponential Almon specification, often called *nealmon*, imposes a smooth lag pattern with only a few parameters. The normalized beta specification, or *nbeta*, is more flexible and can capture hump-shaped lag patterns. This distinction is important in the energy-price application because the effect of crude oil price changes on consumer energy prices may peak after a delay rather than immediately.

Unrestricted MIDAS, discussed by Foroni et al. (2015), estimates each high-frequency lag coefficient separately. This removes the need to choose a parametric lag shape, but it increases the number of coefficients. In small samples, this flexibility can be costly. In this thesis, U-MIDAS is useful as a benchmark because it tests whether the parametric MIDAS restrictions are too strong.

Penalized and non-parametric MIDAS approaches provide intermediate solutions. LASSO-MIDAS applies an L_1 penalty to the lag coefficients, allowing the data to select which weekly lags are most useful. Kernel MIDAS, related to the work of Breitung and Roling (2015), imposes smoothness without forcing a specific parametric shape. These methods help test whether the estimated lag pattern is robust to alternative forms of regularization.

The MIDAS family is especially relevant for this thesis because it spans a meaningful spectrum between structure and flexibility. Parametric MIDAS imposes interpretable smoothness, U-MIDAS allows fully free lag coefficients, and penalized or kernel-based variants sit between these two poles. This makes the MIDAS family a natural testing ground for the broader question of whether forecasting gains come from preserving weekly timing itself, from the shape restrictions imposed on that timing, or from more general model flexibility.

2.5 State-Space and Composite-Link Models

State-space models are another important approach to mixed-frequency data. Mariano and Murasawa (2003) show how variables observed at different frequencies can be combined in a state-space framework. Aruoba et al. (2009) also use state-space methods to measure business conditions in real time. These models are powerful because they can handle latent states, missing observations, and dynamic error structures.

However, many state-space applications focus on large systems or latent factors. This thesis uses a smaller and more transparent setting: monthly consumer energy prices and weekly WTI oil prices. The CLM-SS model is designed to estimate direct weekly link weights while allowing the remaining error to follow an autoregressive process. Its role is not only to compete with MIDAS, but also to test whether a more explicit exact-aggregation framework recovers a similar oil-price transmission pattern. The model is also connected to the broader idea of composite link functions (Thompson & Baker, 1981) and implemented using state-space tools available in R (Helske, 2017).

This is important from a research-design perspective. MIDAS is often attractive because it is parsimonious, while state-space methods are often attractive because they can model dynamic residual structure more explicitly. The CLM-SS model therefore acts as a bridge case in the thesis: it preserves weekly timing and interpretability, but adds a richer time-series error structure. If it performs similarly to MIDAS, that would suggest the core gain comes from the weekly timing itself. If it performs materially better, that would suggest that exact aggregation plus dynamic error modeling adds empirical value beyond standard MIDAS.

2.6 Machine-Learning Benchmarks

Machine-learning methods are increasingly used in forecasting. XGBoost, introduced by Chen and Guestrin (2016), is a flexible tree-based method that can capture nonlinearities and interactions. In this thesis, XGBoost treats weekly WTI lags as tabular features. It is a useful benchmark because it tests whether a flexible nonlinear method can outperform structured mixed-frequency models.

LSTM networks are also included as a deep-learning benchmark. LSTMs are designed for sequential data and can, in principle, learn temporal patterns in weekly oil-price movements. Fischer and Krauss (2018) show that LSTM networks can perform well in financial forecasting settings. However, deep-learning models usually require more data than traditional econometric models. This is a limitation here because the effective sample of monthly target observations remains relatively small, even after extending the data through 2026. For that reason, the LSTM is included mainly as a robustness and defense benchmark rather than as the main modeling approach.

The machine-learning comparison is important for the performance-versus-interpretability trade-off. If XGBoost or LSTM strongly outperformed MIDAS, then the case for interpretable mixed-frequency models would be weaker. More broadly, the comparison helps answer a practical question that motivated the capstone from the beginning: in an applied forecasting setting, is MIDAS still competitive once modern flexible prediction methods are included? This is particularly relevant in small macroeconomic and energy datasets, where regularization and structure can sometimes be more useful than highly flexible nonlinear models (Medeiros et al., 2021).

2.7 Oil Prices and Consumer Energy Inflation

The empirical application is motivated by the relationship between oil prices and consumer energy prices. Hamilton (1983) and Hamilton (2009) show that oil price shocks are important for macroeconomic outcomes, while Kilian (2009) emphasizes that oil price movements can reflect different types of shocks. Baumeister and Hamilton (2019) also highlight the importance of careful structural interpretation in oil-market models.

Consumer energy prices are more directly linked to crude oil prices than broad headline inflation, but the pass-through is not instantaneous. Retail prices depend on refining, distribution, taxes, inventories, and pricing decisions. This creates a natural reason to study timing. Weekly WTI prices may contain information about monthly consumer energy prices, but the relevant information may come from specific weeks rather than from the monthly average. Related work on oil-price forecasting and macroeconomic prediction also supports the idea that crude oil prices contain useful predictive information, although the strength of this relationship can vary over time (Baumeister & Kilian, 2015; Nonejad, 2022).

This is where mixed-frequency models are useful. They allow the thesis to ask not only whether oil prices help forecast consumer energy prices, but also when the oil-price signal is strongest. That timing question is particularly important in the present application because the empirical results are later interpreted in terms of an oil-to-consumer-energy transmission window rather than only in terms of forecast ranking.

2.8 Research Gap

The literature provides several tools for mixed-frequency forecasting, but each approach leaves part of the problem unresolved. ARIMAX and bridge equations are simple, but they rely on pre-aggregation and cannot identify which high-frequency observations matter most. Parametric MIDAS models preserve high-frequency timing, but impose a specific lag-weight shape. U-MIDAS removes this restriction, but may become unstable when the number of lags is large relative to the sample size. State-space models can handle dynamic errors, but many applications focus on large latent-factor systems rather than a transparent bivariate cross-frequency link. Machine-learning methods can be flexible, but they often reduce interpretability and require more data.

This thesis addresses this gap by comparing these approaches in one common empirical setting: forecasting monthly US consumer energy price changes using weekly WTI crude oil price changes. The contribution is not only to ask whether MIDAS improves forecast accuracy, but also whether the improvement remains interpretable and whether it survives outside the original model-development window. In other words, the thesis is concerned with both internal comparison and later validation. This is why the empirical design ultimately combines an initial pseudo-out-of-sample model-development exercise, a post-2022 external holdout, and scenario-based forward projections.

Table 1: Summary of Mixed-Frequency Forecasting Approaches

Approach	Main strength	Main limitation
ARIMAX / bridge equations	Simple and easy to interpret	Require pre-aggregation of high-frequency information
Parametric MIDAS	Preserves high-frequency timing with few parameters	Depends on a chosen lag-weight shape
U-MIDAS	Estimates each high-frequency lag freely	Can be unstable in small samples with many lags
Kernel MIDAS	Allows flexible smooth lag patterns	Less directly interpretable than parametric MIDAS
State-space models	Handle dynamic errors and timing structures	Often require stronger assumptions or larger systems
Machine learning	Flexible nonlinear prediction benchmarks	Lower interpretability and higher data requirements
CLM-SS	Preserves exact weekly link weights and AR errors	More complex to estimate; empirical value must be tested

The research gap can be summarized as follows: existing methods either preserve high-frequency timing, impose interpretable structure, or model time-series error dynamics, but they rarely combine all three in a small-sample applied forecasting setting. The CLM-SS framework is included as an attempt to bridge this gap, while the broader model comparison tests whether this additional structure is empirically competitive with MIDAS, ARIMAX, penalized regression, and machine-learning alternatives. At the same time, the thesis extends the usual mixed-frequency comparison by asking a more applied question: do the same conclusions continue to hold when the models are tested on genuinely later data rather than only within the original development period?

3 Methodology

This chapter explains the empirical design, data sources, model specifications, forecast evaluation strategy, and reproducibility workflow used in the thesis. The methodology is designed to answer the main research question: whether preserving weekly oil-price timing improves forecasts of monthly US consumer energy prices relative to models that first aggregate weekly information to the monthly frequency.

3.1 Research Design

This thesis uses an applied forecasting design to compare monthly aggregation with mixed-frequency alternatives. The goal is to evaluate whether weekly oil price information improves forecasts of monthly US consumer energy prices when the weekly timing is preserved rather than averaged away. The empirical strategy is based on a common forecasting setup: all models use the same target variable, the same predictor design, and comparable forecast metrics.

The main benchmark is a monthly ARIMAX model, which follows the standard time-series logic of modelling a low-frequency outcome with lagged dynamics and an exogenous predictor (Box & Jenkins, 1970). In this benchmark, weekly WTI oil price changes are first aggregated to the monthly frequency. The main alternatives are MIDAS models, which allow high-frequency observations to enter a low-frequency regression directly through estimated lag weights (Ghysels et al., 2004, 2007). The methodology also includes U-MIDAS, CLM-SS, LASSO-MIDAS, Kernel U-MIDAS, XGBoost, and LSTM benchmarks to test whether MIDAS is competitive against natural econometric, state-space, penalized, and machine-learning alternatives (Aruoba et al., 2009; Chen & Guestrin, 2016; Fischer & Krauss, 2018; Foroni et al., 2015; Mariano & Murasawa, 2003).

The final empirical design has three layers. First, the original model-development exercise uses an expanding-window pseudo-out-of-sample design over January 2015 to December 2022. Second, the thesis extends the analysis with a genuinely later external holdout over January 2023 to April 2026. Third, the thesis produces scenario-based true future forecasts beginning in July 2026. This structure allows the models to be compared not only within the original development window, but also on later data and in a forward-looking setting.

3.2 Data Sources and Transformations

The empirical analysis uses publicly available energy-price data. The dependent variable is the US Consumer Energy CPI, obtained from FRED under the code CPIENGSL. This series measures consumer-facing energy prices and is observed monthly. The main predictor is the weekly WTI crude oil price series from FRED, obtained under the code WCOILWTICO. A third series, the IMF Global Price of Energy Index, PNRGINDEXM, is used only for exploratory analysis and motivation. It is not included as a forecasting predictor in the final models.

The original capstone design was conceptually motivated by higher-frequency crude oil data, including daily oil prices. However, the final forecasting workflow uses the official weekly FRED WTI series because it gives a transparent and stable mixed-frequency structure that matches the thesis design more naturally. Each month is represented by four weekly WTI observations, which avoids an excessively large daily lag structure and keeps the models interpretable.

Table 2: Data Used in the Empirical Analysis

Variable	Code	Frequency	Role
Consumer Energy CPI	CPIENGSL	Monthly	Target variable
WTI crude oil price	WCOILWTICO	Weekly	Main high-frequency predictor
IMF Energy Price Index	PNRGINDEXM	Monthly	Exploratory/context variable

All price series are transformed into log-differences. For a price series P_t , the transformed variable is

$$\Delta \log(P_t) = \log(P_t) - \log(P_{t-1}). \quad (6)$$

This transformation is used because the level series are non-stationary, while the log-difference series are stationary and more appropriate for forecasting. Augmented Dickey-Fuller tests on

both the original and refreshed samples support this transformation choice. The original model-development sample runs from February 2000 to December 2022. After refreshing the data, the later evaluation sample extends through April 2026 for CPI and through May 2026 for WTI and the IMF energy index.

For the mixed-frequency design, each month is represented by four weekly WTI observations. This gives a simple and transparent frequency ratio of $m = 4$, where the monthly target depends on weekly oil price information.

3.3 Benchmark Models

The first benchmark is ARIMAX. Weekly WTI changes are averaged into a monthly value,

$$\bar{x}_t = \frac{1}{4} \sum_{j=1}^4 x_{t,j}, \quad (7)$$

where $x_{t,j}$ is the weekly WTI log-change in week j of month t . The ARIMAX model is then estimated as a monthly time-series model with $\bar{x}_t$ as an exogenous regressor:

$$y_t = \alpha + \phi y_{t-1} + \beta \bar{x}_t + \varepsilon_t. \quad (8)$$

This model is useful because it represents the standard pre-aggregation approach. Its limitation is that all weekly observations receive equal weight before estimation. Therefore, ARIMAX answers a simple benchmark question: how well can a monthly model forecast consumer energy prices after the weekly oil-price information has already been collapsed into one monthly value?

The main mixed-frequency models are MIDAS regressions. A MIDAS model can be written as

$$y_t = \alpha + \sum_{k=0}^K w(k; \theta) x_{t-k/4} + \varepsilon_t, \quad (9)$$

where $w(k; \theta)$ is a lag-weight function. Two parametric specifications are used: exponential Almon weights, referred to as *nealmon*, and normalized beta weights, referred to as *nbeta*. These specifications reduce the number of parameters while allowing the model to estimate which weekly lags matter most (Ghysels et al., 2007).

U-MIDAS removes the parametric lag-shape restriction and estimates a separate coefficient for each weekly lag:

$$y_t = \alpha + \sum_{k=0}^K \beta_k x_{t-k/4} + \varepsilon_t. \quad (10)$$

This model is more flexible than parametric MIDAS, but it also uses more parameters. In small monthly samples, this can increase estimation uncertainty (Feroni et al., 2015). U-MIDAS is included to test whether the imposed MIDAS lag shapes are too restrictive.

In the final implementation, the MIDAS-family models use the same weekly lag design so that the models remain directly comparable. Hyperparameter choices such as the number of weekly lags are selected using only the original 2000–2022 sample, before the later external holdout is evaluated. This separation is important because it prevents the later data from being used implicitly in model tuning.

3.4 Composite-Link State-Space Model

The CLM-SS model is included because the original capstone brief focused on exact cross-frequency aggregation. The model links weekly WTI observations to the monthly CPI outcome through a composite link matrix Z_t . The observation equation is

$$y_t = \mu + Z_t w + u_t, \quad (11)$$

where Z_t contains the weekly WTI observations assigned to month t , w is the vector of estimated weekly link weights, and u_t is the remaining time-series error. The error follows an AR(1) process:

$$u_t = \rho u_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma^2). \quad (12)$$

This model is connected to the broader state-space literature on mixed-frequency data, where high- and low-frequency information can be combined within a single dynamic system (Aruoba et al., 2009; Helske, 2017; Mariano & Murasawa, 2003). In this thesis, the CLM-SS model is kept relatively transparent. Instead of estimating a large latent-factor system, it estimates direct weekly link weights between WTI oil changes and monthly consumer energy price changes.

The parameters are estimated by maximum likelihood using BFGS optimization. The model is checked for identifiability using the Hessian matrix and standard errors. A ridge-penalized version is also tested as a sensitivity check. The purpose of CLM-SS is not only to compete with MIDAS in forecast accuracy, but also to see whether a direct exact-aggregation structure recovers a similar weekly lag pattern.

3.5 Additional Benchmarks

Several additional models are included for robustness. LASSO-MIDAS applies an L_1 penalty to the weekly lag coefficients, allowing the model to shrink less useful lags toward zero. Penalized forecasting methods are useful in settings where there are many possible predictors relative to the sample size (Medeiros et al., 2021). Kernel U-MIDAS adds a smoothness penalty to unrestricted lag coefficients, following the idea that nearby high-frequency lags may have related effects (Breitung & Roling, 2015).

XGBoost is used as a nonlinear tree-based benchmark (Chen & Guestrin, 2016). It treats weekly WTI lags as tabular features and can capture nonlinear effects and interactions. LSTM is used as a recurrent neural-network benchmark because it is designed for sequential data (Fischer & Krauss, 2018). These machine-learning models help answer whether more flexible prediction methods outperform structured mixed-frequency models. They are also useful for the performance-versus-interpretability comparison, because strong machine-learning performance would weaken the case for using more interpretable MIDAS-type models.

Principal-components compression and VIF diagnostics are also included as robustness checks. These additional tests help determine whether any MIDAS advantage is genuinely due to weekly timing information rather than simply to multicollinearity, dimensionality, or an overly rich lag matrix.

3.6 Forecast Evaluation

Forecast performance is evaluated in stages.

The first stage is the original expanding-window pseudo-out-of-sample design. The initial training period runs from February 2000 to December 2014. The first test period runs from January 2015

to December 2022, giving 96 one-step-ahead forecasts. After each forecast, the training sample expands by one month.

The second stage is the post-2022 external holdout. In this design, the first later forecast is made for January 2023 using only information available through December 2022. The sample then expands month by month through April 2026, producing 38 additional one-step-ahead forecasts. This later holdout is important because it tests whether conclusions drawn from the original model-development period remain valid on genuinely later data.

The third stage is a set of true future scenario forecasts. These begin in July 2026. The main thesis reports short-horizon projections for 1, 3, and 6 months, while a 12-month scenario extension is reserved for the appendix. These forecasts are conditioned on assumed oil-price paths rather than evaluated against known future CPI values.

The main accuracy metrics are RMSE and MAE:

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2}, \quad (13)$$

$$MAE = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t|. \quad (14)$$

The thesis also reports MASE and sMAPE as scale-free metrics. MAPE is reported with caution because the dependent variable is a monthly log-change and can be close to zero. When the actual value is near zero, percentage errors can become very large even if the absolute forecast error is not especially large. For this reason, MAPE is treated as a secondary diagnostic rather than as the main model-ranking metric.

Directional accuracy is used to evaluate whether the model predicts the correct sign of the monthly change:

$$DA = \frac{1}{T} \sum_{t=1}^T \mathbf{1} [\text{sign}(\hat{y}_t) = \text{sign}(y_t)]. \quad (15)$$

Large-move precision and recall are also reported. These metrics evaluate whether the models capture months with unusually large consumer energy price changes. Precision measures how often predicted large moves are actually large moves, while recall measures how many actual large moves are detected by the model. This is useful because energy-price forecasting is not only about minimizing average error, but also about identifying periods of stress.

Diebold-Mariano tests are used to compare forecast errors against the ARIMAX benchmark (Diebold & Mariano, 1995). Mincer-Zarnowitz regressions are also used to evaluate forecast rationality (Mincer & Zarnowitz, 1969):

$$y_t = \alpha + \beta \hat{y}_t + e_t. \quad (16)$$

In an ideal unbiased forecast, $\alpha = 0$ and $\beta = 1$. In this thesis, the Mincer-Zarnowitz test is interpreted together with predictive R^2 , because a model can appear unbiased while still having weak predictive power.

For the true future scenario forecasts, the interpretation is different. Since realized CPI values are not yet available, these projections are not evaluated using forecast errors. Instead, they are read as model-implied conditional paths under alternative oil assumptions. The short-horizon scenarios use flat, recent-mean, upside-shock, and downside-shock oil paths; full scenario tables and the 12-month extension are reported in Appendix D.

3.7 Reproducibility

The full empirical workflow is implemented in R and is publicly available at https://github.com/stefhooy/midas_capstone. Figures and tables are saved in the `output/figures` and `output/tables` folders, processed intermediate objects are saved in `data/processed`, and a fixed random seed is used where relevant. Tuning decisions are selected using only the original 2000–2022 sample before the later post-2022 holdout is evaluated. A complete list of scripts and implementation details is provided in Appendix E.

4 Results

This section presents the main empirical findings. Out-of-sample forecast accuracy is reported first for the model-development window and the post-2022 external holdout. A short interpretability subsection summarizes the oil-to-energy transmission window, and a robustness subsection synthesizes the supplementary diagnostic evidence reported in Appendix C. Additional forecast tables, weight plots, and scenario projections appear in Appendices A–D.

4.1 Out-of-Sample Forecast Accuracy

4.1.1 Model-Development Window: 2015–2022

The original model-development evaluation is based on an expanding-window forecasting exercise. The initial training period runs from February 2000 to December 2014, and the test period runs from January 2015 to December 2022. This gives 96 one-step-ahead forecasts for each model.

Table 3 reports the main out-of-sample accuracy metrics for this original development window. RMSE is the main loss measure, while MAE and MASE provide robustness checks. MAPE and sMAPE are also reported, but MAPE should be interpreted cautiously because the dependent variable is a monthly log-change and can be close to zero.

Table 3: Out-of-Sample Forecast Accuracy, January 2015–December 2022

Rank	Model	RMSE	MAE	MASE	MAPE	sMAPE	Dir. Acc.	vs ARIMAX RMSE
1	XGBoost	0.02170	0.01559	0.562	117.3%	100.6%	79.2%	-21.3%
2	LASSO-MIDAS	0.02265	0.01694	0.610	125.7%	104.1%	76.0%	-17.9%
3	LSTM	0.02505	0.01821	0.656	158.9%	103.5%	71.9%	-9.2%
4	PCA-ARIMAX	0.02663	0.01901	0.685	127.3%	103.9%	77.1%	-3.4%
5	Kernel U-MIDAS	0.02705	0.01836	0.662	149.2%	99.9%	77.1%	-1.9%
6	U-MIDAS	0.02711	0.01787	0.644	138.1%	96.1%	78.1%	-1.7%
7	CLM-SS (12 lags)	0.02750	0.01850	0.667	145.3%	98.2%	76.0%	-0.3%
8	ARIMAX	0.02758	0.02058	0.742	125.1%	125.0%	64.6%	baseline
9	MIDAS nbeta	0.02913	0.01888	0.680	147.1%	99.4%	78.1%	+5.6%
10	CLM-SS (4 lags)	0.03187	0.02176	0.784	132.4%	129.2%	66.7%	+15.5%
11	MIDAS nealmon	0.03410	0.01993	0.718	164.4%	95.6%	81.2%	+23.6%

The original 2015–2022 results do not support a simple one-model-wins-everywhere conclusion. XGBoost achieves the lowest RMSE in this development window, followed by LASSO-MIDAS. U-MIDAS, Kernel U-MIDAS, and CLM-SS (12 lags) all perform roughly on par with ARIMAX, while the two parametric MIDAS specifications are more mixed: nbeta is competitive in directional terms but not in RMSE, while nealmon has the highest directional accuracy but the weakest RMSE. In-sample information criteria reported in Appendix A favor more flexible specifications, but these rankings do not carry over directly to the out-of-sample exercise.

4.1.2 Post-2022 External Holdout: 2023–2026

The later external holdout is the strongest validation exercise in the thesis. In this design, the first forecast is made for January 2023 using only information available through December 2022. The

window then expands month by month through April 2026, yielding 38 one-step-ahead forecasts on genuinely later data.

Table 4 reports the main forecast metrics for this holdout period.

Table 4: Post-2022 External Holdout Forecast Accuracy, January 2023–April 2026

Rank	Model	RMSE	MAE	MASE	sMAPE	Dir. Acc.	vs ARIMAX RMSE	DM p-value
1	MIDAS nbeta	0.01622	0.01234	0.456	127.0%	57.9%	-16.0%	0.1443
2	U-MIDAS	0.01639	0.01286	0.476	114.8%	71.1%	-15.1%	0.2242
3	LASSO-MIDAS	0.01703	0.01295	0.479	120.2%	71.1%	-11.9%	0.2091
4	MIDAS nealmon	0.01762	0.01272	0.470	129.8%	63.2%	-8.8%	0.1610
5	CLM-SS (12 lags)	0.01825	0.01377	0.509	109.7%	71.1%	-5.5%	0.3994
6	LSTM	0.01860	0.01310	0.484	118.3%	63.2%	-3.7%	0.3485
7	XGBoost	0.01929	0.01369	0.506	119.8%	65.8%	-0.1%	0.4931
8	ARIMAX	0.01932	0.01247	0.461	122.9%	65.8%	baseline	–

On the cleaner post-2022 holdout, MIDAS with normalized beta weights becomes the best-performing model by RMSE, reducing forecast error by 16.0% relative to ARIMAX. Diebold–Mariano tests do not reject equal forecast accuracy against ARIMAX at conventional levels, which reflects the short holdout length as much as the absence of any improvement. U-MIDAS and LASSO-MIDAS are very close behind, and MIDAS nealmon clearly improves relative to ARIMAX. By contrast, XGBoost, which was the best model in the original 2015–2022 development window, is essentially tied with ARIMAX on the later holdout.

This is a strong result for the thesis because it shows that the weekly-timing advantage survives on genuinely later data. Over the broader combined 2015–2026 window, XGBoost again ranks first on RMSE, while MIDAS nbeta ranks first in the later holdout (Appendix A). Model rankings therefore depend on the evaluation frame rather than on a single universal winner.

4.2 Interpretability: Oil-to-Energy Transmission Window

Beyond forecast accuracy, the thesis asks whether the models recover an economically meaningful timing pattern. Across MIDAS, CLM-SS, LASSO-MIDAS, Kernel U-MIDAS, and XGBoost, the strongest oil-price signal appears around prior-month weeks 2–3. This implies an approximate 5–7 week transmission window from weekly crude oil price movements to monthly consumer energy price changes.

Figure 6 in Appendix B shows that the normalized beta MIDAS specification can accommodate a hump-shaped lag profile concentrated in this delayed weekly window. Figure 7 in the same appendix shows that the 12-lag CLM-SS model independently recovers a similar pattern without imposing the same parametric lag shape. CLM-SS does not consistently beat the strongest forecasting models in RMSE terms, but it strengthens the interpretation that the delayed transmission pattern is not simply an artifact of one MIDAS weighting function.

4.3 Robustness Summary

Several supplementary checks support the main forecasting conclusions; full figures and tables are reported in Appendix C.

First, multicollinearity is not a concern: all VIF values for weekly WTI lags remain far below conventional thresholds (Table 10). Second, dimensionality reduction alone does not explain the mixed-frequency gains: PCA-ARIMAX improves only modestly on ARIMAX and remains weaker than the stronger weekly-timing models (Figure 11). Third, directional and large-move diagnostics show that MIDAS nealmon is especially strong at sign prediction in the development window, while weekly-timing models achieve relatively high recall for large CPI Energy moves (Figures 8 and 9). Fourth, the mixed-frequency advantage is strongest at short forecast horizons and remains

visible across pre-COVID, COVID, and recovery periods, although the gains are largest in volatile episodes (Tables 7, 8, and 9). Finally, Mincer–Zarnowitz regressions show that ARIMAX appears closer to the formal unbiased benchmark, but this partly reflects its weaker predictive variation rather than superior forecast usefulness (Figure 10).

The thesis also extends beyond observed data through scenario-based future projections beginning in July 2026. These conditional paths are not validated out-of-sample forecasts; they illustrate how alternative oil-price assumptions translate into different CPI Energy projections under each model class (Appendix D).

5 Discussion

This chapter interprets the empirical results and connects them back to the research question. The main conclusion is that preserving weekly oil-price timing improves forecasting performance relative to simple monthly aggregation, but the strength of that advantage depends on the evaluation setting. The discussion therefore focuses on three main ideas: why weekly timing matters, why the model rankings differ across evaluation windows, and what the results imply for interpretability, forecast use, and future work.

5.1 Why Weekly Timing Matters

The core reason mixed-frequency models can outperform ARIMAX is that ARIMAX pre-aggregates weekly oil-price information before estimation. In the monthly benchmark, the four weekly WTI observations inside a month are averaged into a single monthly value. This assumes that all weeks are equally informative. The empirical results show that this assumption is too restrictive.

The weekly-timing models preserve the internal structure of the predictor. Instead of assigning equal importance to all weeks in advance, they allow the data to determine how the effect of weekly WTI changes is distributed over time. This matters because the pass-through from crude oil prices to consumer energy prices is not immediate. The estimated lag patterns repeatedly show that the strongest signal appears around prior-month weeks 2–3, corresponding to an approximate 5–7 week transmission window. A monthly average cannot recover that timing pattern.

This result is also economically plausible. Consumer energy prices do not react to crude oil movements instantaneously. Pass-through can be delayed by refining, inventories, transportation, contracts, taxes, and pricing decisions. The forecasting advantage of weekly-timing models is therefore not only a statistical artifact. It reflects the fact that oil-price information reaches consumer-facing energy prices with a short but meaningful delay.

5.2 Why the Model Rankings Change Across Evaluation Windows

One of the most important findings of the thesis is that the model rankings depend on the evaluation frame. In the original 2015–2022 model-development window, the strongest RMSE performance comes from XGBoost, followed by LASSO-MIDAS. In that period, the parametric MIDAS specifications are more mixed: MIDAS nbeta is not the best RMSE model, while MIDAS nealmon performs especially well in directional accuracy but poorly in RMSE.

However, the interpretation changes in the later 2023–2026 external holdout. In that genuinely later period, MIDAS nbeta becomes the best model by RMSE, reducing forecast error by 16.0% relative to ARIMAX. U-MIDAS and LASSO-MIDAS are close behind, while XGBoost is no longer meaningfully better than ARIMAX.

This does not weaken the thesis. On the contrary, it improves it. If one model had won every table by a large margin, the result would have been simpler but not necessarily more informative. The fact that the rankings change tells us something useful about model behavior. The original

2015–2022 exercise is best understood as a model-development and comparison period. The later holdout is a cleaner validation test. In that cleaner test, the weekly-timing advantage survives, and it is MIDAS nbeta that performs best.

The broader combined 2015–2026 summary adds one more layer. Across the full combined period, XGBoost again has the lowest RMSE, with LASSO-MIDAS close behind. This suggests that XGBoost is a strong long-period forecasting benchmark, but it does not overturn the later holdout result. A reasonable interpretation is that the machine-learning models perform well when judged over a broad mixed environment, while MIDAS nbeta is especially strong when tested on genuinely later data outside the original development window.

5.3 Why MIDAS nbeta is Selected as the Main Model

The final choice of a main model depends on what the thesis is trying to emphasize. If the objective were only to identify the single lowest RMSE in the original 2015–2022 development sample, then XGBoost would deserve that role. But that would not fully answer the capstone question, because the thesis is not only about model competition inside one historical window. It is also about whether preserving weekly timing remains useful when the models are taken outside the original development period.

For that reason, MIDAS nbeta is selected as the main model. It is the best-performing model in the post-2022 external holdout, which is the strongest later validation exercise in the thesis. It is also highly interpretable, since its lag-weight structure can be directly linked to an economically meaningful transmission window. In addition, among the parametric MIDAS specifications it performs better than nealmon in the in-sample information criteria.

MIDAS nealmon remains important, however, and should not be treated as a failed model. In the original 2015–2022 exercise, it achieves the highest directional accuracy, correctly predicting the sign of monthly CPI Energy changes in 81.2% of the test months. This makes nealmon especially useful if the practical objective is sign prediction rather than exact magnitude prediction. The most balanced interpretation is therefore that MIDAS nbeta is the preferred main specification for the thesis, while MIDAS nealmon remains the strongest directional benchmark within the parametric MIDAS family.

5.4 What the Precision and Recall Results Mean

The directional and large-move results add nuance to the RMSE rankings. Across the main weekly-timing models, recall for large movements is relatively strong, while precision is more moderate. This means the models are good at detecting most of the important energy-price moves, but they also generate some false alarms.

This trade-off is understandable in the context of energy prices. Large monthly changes in Consumer Energy CPI are relatively rare, while oil prices are much more volatile. A model that reacts to oil-price information may sometimes flag a large movement in consumer energy prices that does not fully materialize. That lowers precision. At the same time, missing a genuinely important energy-price move may be more costly than issuing a false warning, especially in an applied risk or monitoring context. In that sense, high recall can still be valuable even when precision is not especially high.

This is one reason the thesis does not rely on a single metric. RMSE, MAE, and MASE measure average forecast error; directional accuracy measures sign prediction; recall measures whether important large moves are detected. Looking at these metrics together gives a more complete view of forecast usefulness.

5.5 The Mincer-Zarnowitz Paradox

The Mincer-Zarnowitz results create an important interpretation issue. ARIMAX appears closer to the formal unbiased benchmark in the Mincer-Zarnowitz regressions, but its predictive content is weak. This means that ARIMAX can look statistically cleaner partly because it does not move strongly enough to explain much variation in the target variable.

The weekly-timing models often show the opposite pattern. They depart more from the textbook ideal in the Mincer-Zarnowitz regression, but they contain more useful forecast variation and reduce forecast error more meaningfully. This creates what can be described as a Mincer-Zarnowitz paradox: the apparently cleaner model is not the more useful forecasting model.

In this thesis, the Mincer-Zarnowitz test is therefore interpreted jointly with forecast accuracy and predictive strength rather than in isolation. A forecast should not be preferred only because it appears more formally unbiased if its predictive value is limited. The mixed-frequency models are more useful because they contain more informative variation, even if they are not perfectly calibrated in the narrow Mincer-Zarnowitz sense.

5.6 Forecast Horizon Validity

The main forecasting exercise is best interpreted as a short-horizon mixed-frequency forecasting or nowcasting exercise. At horizon $h = 1$, the model uses weekly WTI information that is available before the monthly CPI release and maps it into the monthly CPI Energy outcome. This is a realistic mixed-frequency setting, because the higher-frequency oil-price information arrives before the lower-frequency target is published.

The forecast-horizon evidence suggests that the value of weekly timing is strongest at short horizons. In the original horizon-validity exercise, the relative performance of the MIDAS specifications changes as the horizon increases, and the advantage becomes less stable beyond the near term. This is consistent with the estimated 5–7 week transmission window: once the forecast horizon moves too far beyond that window, weekly oil-price information becomes less informative for monthly consumer energy price changes.

This is also why the true future forecasting section of the thesis is organized around 1-, 3-, and 6-month scenario windows, with the 12-month extension moved to the appendix. The short-horizon projections are aligned with the economic timing found in the model weights, while the longer-horizon scenario extension is better treated as a conditional exploratory exercise rather than as the core forecast evidence.

5.7 Interpretation of CLM-SS

The CLM-SS model was motivated by the original capstone brief, which focused on exact cross-frequency aggregation. In the empirical results, the short-lag CLM-SS specification performs poorly, while the 12-lag version is much more competitive. This confirms that the relevant oil-price information extends beyond the current month.

CLM-SS does not emerge as the top forecasting model, but it still plays an important role in the thesis. First, it shows that a more explicit exact-aggregation framework can produce competitive forecasts when enough weekly information is included. Second, and more importantly, it recovers a delayed transmission pattern that is broadly consistent with the MIDAS evidence. This strengthens the interpretation that the lag hump is real and not simply an artifact of one specific MIDAS parameterization.

The AR error component in CLM-SS appears to add some value, but not enough to dominate the simpler mixed-frequency alternatives. One possible interpretation is that once enough weekly

oil-price lags are included, much of the relevant persistence in monthly consumer energy inflation is already being captured. In that sense, CLM-SS acts more as a confirmatory structural bridge between ARIMAX and MIDAS than as the final forecasting winner.

5.8 What the Machine-Learning Results Mean

The machine-learning results should also be interpreted carefully. XGBoost performs very well in the original development window and in the broader combined 2015–2026 robustness summary. This shows that flexible nonlinear methods can be highly competitive in this forecasting problem. LASSO-MIDAS also performs strongly, which suggests that some of the useful signal can be recovered through regularized lag selection.

However, the later external holdout changes the emphasis. In that later period, XGBoost is no longer the best model, while MIDAS nbeta, U-MIDAS, and LASSO-MIDAS outperform the monthly benchmark more clearly. This suggests that structured mixed-frequency models may generalize more robustly to later data in this application. In other words, machine-learning flexibility is clearly valuable, but it does not remove the case for interpretable mixed-frequency structure.

This is one of the most useful broader lessons from the capstone. The real empirical choice is not between “traditional econometrics” and “machine learning” in the abstract. Instead, the results suggest that different forms of structure, flexibility, and regularization matter differently depending on the evaluation window. The best models are the ones that combine useful signal extraction with enough interpretability to explain why the forecasts move.

5.9 Limitations

Several limitations should be acknowledged. First, the sample remains relatively small for the most flexible models. Even after extending the evaluation through 2026, the number of monthly target observations is still modest for deep-learning methods such as LSTM. The weaker and less stable LSTM performance may therefore reflect data limitations rather than the full potential of the architecture.

Second, the empirical application focuses only on consumer energy prices and WTI crude oil. This is a strength because the economic channel is relatively clear, but it also limits generalizability. The results should not automatically be interpreted as applying to headline CPI, core inflation, GDP, or other macroeconomic outcomes.

Third, the analysis uses a simplified weekly alignment with four weekly WTI observations per month. This makes the mixed-frequency structure transparent and suitable for MIDAS, but it is still an approximation. Daily oil-price data could, in principle, provide a richer high-frequency design, although that would also increase dimensionality and make the lag structure harder to interpret.

Fourth, some robustness components remain relatively simple. The CLM-SS ridge penalty is selected through a practical validation procedure rather than through a larger cross-validation design. The Kernel U-MIDAS smoother is implemented as a practical benchmark rather than as an exhaustive non-parametric exercise. The future scenario forecasts also depend on assumed oil-price paths, which means they should be interpreted as conditional projections rather than as validated long-horizon forecasts.

Finally, MAPE is not emphasized because the dependent variable is a log-change that can be close to zero. In this setting, percentage errors can become unstable and potentially misleading. This is why the thesis places more weight on RMSE, MAE, MASE, sMAPE, directional accuracy, and large-move performance.

5.10 Summary

Overall, the discussion supports the main conclusion of the thesis. Weekly oil-price timing matters because it contains information that is lost through monthly aggregation. The evidence is not that one model wins every table, but that preserving weekly timing repeatedly improves the forecasting design and yields an interpretable 5–7 week transmission window.

MIDAS nbeta is selected as the main model because it performs best in the later external hold-out and remains highly interpretable. MIDAS nealmon remains important because it is particularly strong in directional forecasting. XGBoost and LASSO-MIDAS show that flexible and regularized methods are also strong competitors, especially in the original development window and in the broader combined robustness summary. CLM-SS contributes by confirming the delayed transmission structure through a different exact-aggregation framework.

Taken together, the results suggest that weekly WTI oil-price movements contain useful short-run information for monthly US consumer energy prices, and that mixed-frequency models are most valuable when the forecasting problem requires both timing sensitivity and interpretability.

6 Conclusion

This thesis examined whether mixed-frequency models can improve forecasts of monthly US consumer energy prices by preserving weekly oil-price information. The original motivation came from the broader problem of exact cross-frequency aggregation: many economic outcomes are observed at a lower frequency than their predictors, and standard pre-aggregation can lose useful timing information. The final empirical application focused on monthly US Consumer Energy CPI and weekly WTI crude oil prices because this setting has a clear economic transmission channel, publicly available data, and a natural mixed-frequency structure.

The main research question was whether mixed-frequency models that preserve weekly oil-price timing improve forecasts relative to models that pre-aggregate weekly oil data. The answer is yes, but with an important qualification. The results do not show that one single model dominates in every period and under every evaluation design. Instead, they show that preserving weekly timing repeatedly improves the forecasting framework and becomes especially valuable when the models are tested on genuinely later data.

In the original 2015–2022 model-development exercise, the strongest RMSE performance comes from XGBoost, followed by LASSO-MIDAS. This shows that flexible machine-learning and penalized methods can be highly competitive in this setting. However, the later external holdout from January 2023 to April 2026 provides the stronger validation test. In that later period, MIDAS with normalized beta weights becomes the best-performing model by RMSE, reducing forecast error by 16.0% relative to ARIMAX. This is the most important forecasting result of the thesis, because it shows that the weekly-timing advantage survives outside the original development window.

The first subquestion asked what the natural competitors to MIDAS are in this setting. The thesis compared MIDAS against monthly ARIMAX, U-MIDAS, CLM-SS, LASSO-MIDAS, Kernel U-MIDAS, PCA-ARIMAX, XGBoost, and LSTM. This comparison shows that MIDAS is not only better than a weak benchmark. It remains competitive against unrestricted, penalized, state-space, and machine-learning alternatives, and in the later external holdout it becomes the strongest model.

The second subquestion asked whether MIDAS is competitive enough to be useful in a real forecasting setting. The answer is also yes. The results suggest that structured mixed-frequency models are especially valuable when the goal is to preserve economically meaningful timing information rather than only to maximize flexibility. Even when other methods perform strongly, simple monthly aggregation is not the winner. This means that the mixed-frequency design itself contributes useful forecasting information.

The third subquestion asked whether the models recover an interpretable timing pattern. The an-

swer is clearly yes. Across MIDAS, CLM-SS, LASSO-MIDAS, Kernel U-MIDAS, and XGBoost, the strongest oil-price signal appears around prior-month weeks 2–3. This implies an approximate 5–7 week transmission window from crude oil prices to consumer energy prices. This is one of the most important findings of the thesis because it shows that the better-performing models are not only forecasting more accurately; they are also identifying an economically meaningful lag structure.

The fourth subquestion asked whether the result is specific to MIDAS or partly driven by the dataset. The evidence suggests that both matter. MIDAS performs well because it is designed for mixed-frequency data and can preserve weekly timing. At the same time, the energy-price dataset is especially suitable for this type of model because crude oil prices and consumer energy prices have a strong economic connection and a plausible delayed pass-through mechanism. The conclusion should therefore not be that MIDAS will always dominate in every dataset. A more careful conclusion is that MIDAS performs particularly well when the high-frequency predictor contains useful timing information for the low-frequency target.

The CLM-SS model also contributes to the thesis, even though it does not emerge as the top forecasting model. CLM-SS acts as an interpretable bridge between ARIMAX and MIDAS. Like ARIMAX, it remains transparent and relatively easy to interpret. Like MIDAS, it preserves weekly timing instead of averaging the predictor before estimation. The stronger CLM-SS specification recovers a similar lag structure to the MIDAS models, which supports the original capstone idea that exact cross-frequency aggregation can be useful even when it is not the final forecasting winner.

The thesis also extends beyond historical forecast evaluation through scenario-based future projections. The main forward-looking exercise reports 1-, 3-, and 6-month forecasts from July 2026 under alternative oil-price assumptions, while a 12-month scenario extension is included in the appendix. These projections are not used as proof of forecast superiority, because their realizations are not yet observed. Instead, they show how different mixed-frequency and benchmark models translate assumed oil-price paths into possible future CPI Energy paths. This adds an applied forward-looking dimension to the capstone while keeping the core empirical claims grounded in validated forecast evidence.

Several limitations should be acknowledged. First, the empirical setting uses weekly WTI prices and monthly consumer energy prices. This is a meaningful mixed-frequency structure, but it is not an extreme frequency gap. In a future extension, it would be interesting to study a much higher-frequency predictor, such as daily oil prices or even intraday financial energy prices, against a monthly or quarterly macroeconomic target. A daily dataset could capture richer volatility patterns and make the exact aggregation problem more demanding.

Second, the thesis focuses on one main empirical application. The energy-price setting is useful because the economic mechanism is relatively clear, but the results may not generalize automatically to other mixed-frequency problems. Future work could test the same framework on other datasets, such as daily financial indicators and quarterly GDP, hourly electricity demand and weather variables, or monthly inflation with daily commodity prices.

Third, the CLM-SS model could be developed further. The ridge penalty was selected through a practical validation procedure rather than through a more extensive cross-validation design. The model could also be extended to include multiple high-frequency predictors simultaneously, moving the framework closer to a multivariate mixed-frequency system.

Fourth, machine-learning methods could be explored further with richer data. The LSTM benchmark is informative, but the sample is still relatively small for deep learning. With daily predictors, longer samples, or panel data across several energy components, neural-network methods may become more competitive.

A final direction for future work is to turn the empirical workflow into a reusable R package.

The project already produces a documented R codebase with separate scripts for data preparation, MIDAS benchmarks, CLM-SS estimation, machine-learning benchmarks, robustness checks, refreshed-data evaluation, and future scenario forecasting. Packaging the CLM-SS estimator and the mixed-frequency evaluation tools would make the framework easier to apply to other datasets.

Overall, the thesis shows that preserving high-frequency timing can improve low-frequency forecasts when that timing carries economic meaning. In this application, weekly oil prices contain useful short-run information for monthly US consumer energy prices, and mixed-frequency models are able to extract it. The final evidence is therefore not that one model wins every comparison, but that the mixed-frequency approach itself is empirically valuable. MIDAS with normalized beta weights provides the clearest combination of later holdout performance and interpretability, while the broader comparison also shows the relevance of regularized and machine-learning benchmarks. Taken together, the results support the conclusion that mixed-frequency forecasting is most useful when it combines predictive performance with an interpretable economic timing structure.

A Supplementary Forecast Tables

A.1 In-Sample Fit and Information Criteria

Table 5 reports the in-sample AIC, BIC, and RMSE for the likelihood-based models where information criteria are applicable. AIC and BIC are useful because they balance in-sample fit against model complexity. They do not replace out-of-sample forecast evaluation, but they provide an additional check on whether the better forecasting models are also reasonable from an in-sample model-selection perspective.

Table 5: In-Sample AIC, BIC, and RMSE for Applicable Models

Model	N	Params	AIC	BIC	RMSE
ARIMAX monthly mean	275	4	-1160.00	-1141.92	0.02883
MIDAS nealmon (12 weekly lags)	273	4	-1195.59	-1177.54	0.02660
MIDAS nbeta (12 weekly lags)	273	4	-1205.37	-1187.32	0.02612
U-MIDAS (12 weekly lags)	273	13	-1264.13	-1213.60	0.02270
CLM-SS (12 weekly lags)	273	15	-1266.41	-1212.27	–

The in-sample results are informative, but they need to be interpreted carefully. Among the parametric MIDAS specifications, normalized beta weights perform better than exponential Almon weights on both AIC and BIC. U-MIDAS and CLM-SS achieve even lower AIC values, reflecting their greater flexibility, and U-MIDAS also has the lowest in-sample RMSE. However, these models use more parameters, so their stronger in-sample fit does not automatically imply better forecasting performance.

AIC and BIC are not reported for XGBoost, LSTM, LASSO-MIDAS, Kernel U-MIDAS, or PCA-ARIMAX because these are machine-learning, penalized, or diagnostic models without directly comparable likelihood-based information criteria.

A.2 Updated Combined Robustness Summary: 2015–2026

To complement the separate window analyses, Table 6 combines the original 2015–2022 pseudo-out-of-sample forecasts with the 2023–2026 external holdout, giving 134 one-step-ahead forecasts in total.

Table 6: Updated Combined Out-of-Sample Metrics, January 2015–April 2026

Rank	Model	RMSE	MAE	MASE	sMAPE	Dir. Acc.	vs ARIMAX RMSE
1	XGBoost	0.02105	0.01505	0.557	106.0%	75.4%	-17.5%
2	LASSO-MIDAS	0.02121	0.01581	0.585	108.7%	74.6%	-16.9%
3	LSTM	0.02340	0.01676	0.620	107.7%	69.4%	-8.3%
4	U-MIDAS	0.02455	0.01645	0.609	101.4%	76.1%	-3.8%
5	CLM-SS (12 lags)	0.02522	0.01716	0.635	101.5%	74.6%	-1.1%
6	ARIMAX	0.02551	0.01828	0.676	124.4%	64.9%	baseline
7	MIDAS nbeta	0.02612	0.01702	0.630	107.2%	72.4%	+2.4%
8	MIDAS nealmon	0.03035	0.01789	0.662	105.3%	76.1%	+19.0%

Figure 4 visualizes this broader ranking.

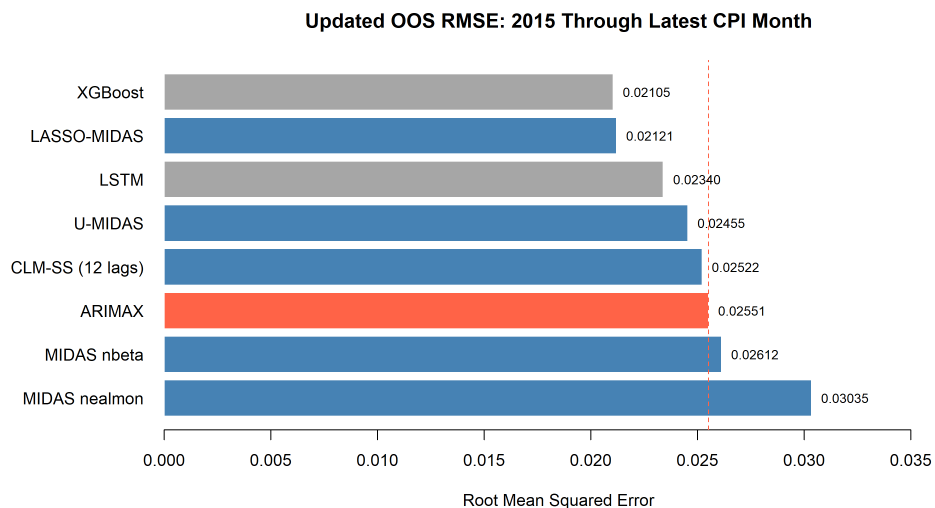


Figure 4: Updated combined RMSE ranking for January 2015 to April 2026. XGBoost is the best model over the full combined period, but the later external holdout separately favors MIDAS nbeta.

A.3 Forecast Paths in the Original Window

Figure 5 compares ARIMAX, MIDAS nealmon, MIDAS nbeta, and U-MIDAS against the realized monthly Consumer Energy CPI log-change in the 2015–2022 development window.

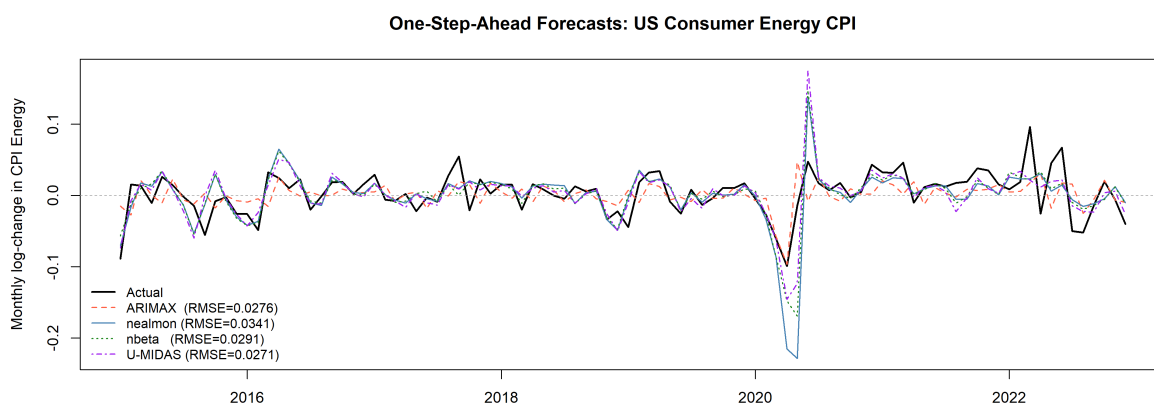


Figure 5: One-step-ahead forecasts of monthly US Consumer Energy CPI log-changes, January 2015 to December 2022. The figure compares ARIMAX, MIDAS nealmon, MIDAS nbeta, and U-MIDAS against the realized series.

B MIDAS and CLM-SS Weight Estimates

B.1 MIDAS Lag Weights

Figure 6 shows the estimated MIDAS lag weights. The key pattern is that the strongest weights appear around prior-month weeks 2–3. This corresponds to an approximate 5–7 week transmission window from weekly crude oil price movements to monthly consumer energy prices.

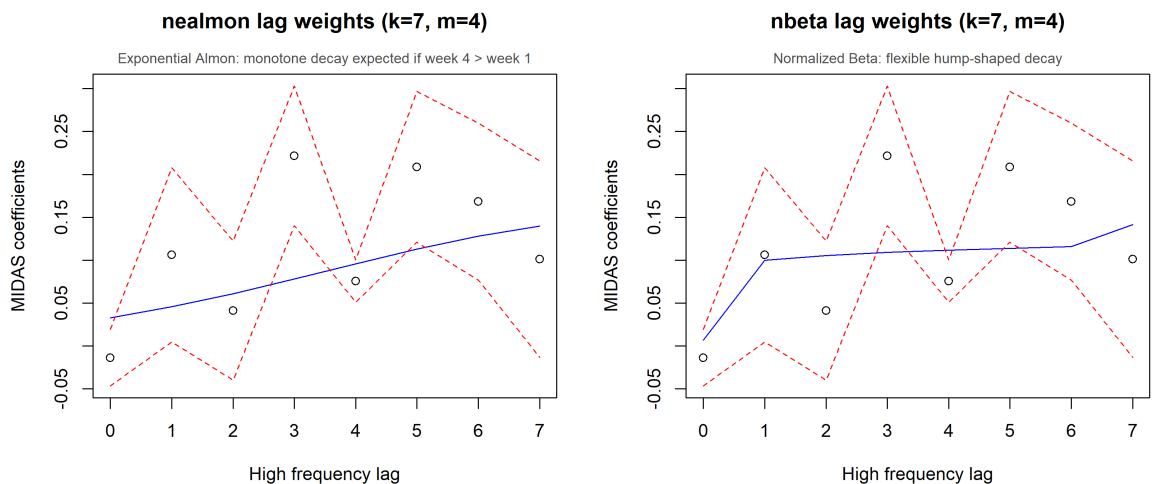


Figure 6: Estimated MIDAS lag weights for the exponential Almon and normalized beta specifications. The lag pattern suggests that oil-price movements from prior-month weeks 2–3 are especially informative for monthly consumer energy price changes.

B.2 CLM-SS Link Weights

Figure 7 reports the CLM-SS composite link weights. The 12-lag CLM-SS model independently recovers a similar pattern to the MIDAS models, with relatively strong weights in the delayed weekly window.

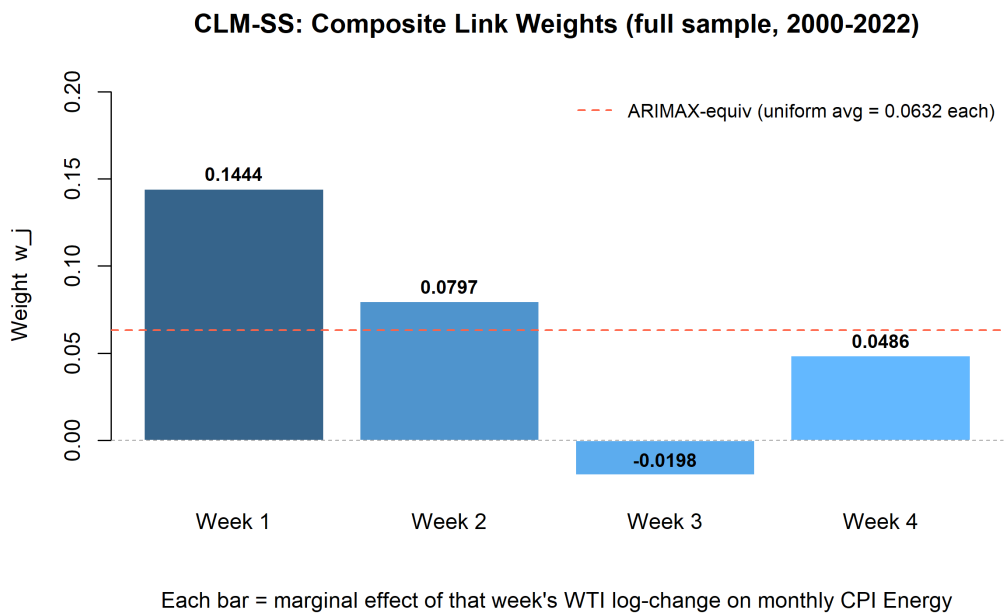


Figure 7: Composite link weights from the CLM-SS model. The estimated weights provide an interpretable weekly aggregation structure and support the same delayed oil-to-consumer-price transmission pattern found by MIDAS.

C Diagnostic and Robustness Results

C.1 Directional, Large-Move, and Rationality Diagnostics

Figure 8 reports directional accuracy across the main benchmark models in the original 2015–2022 evaluation window.

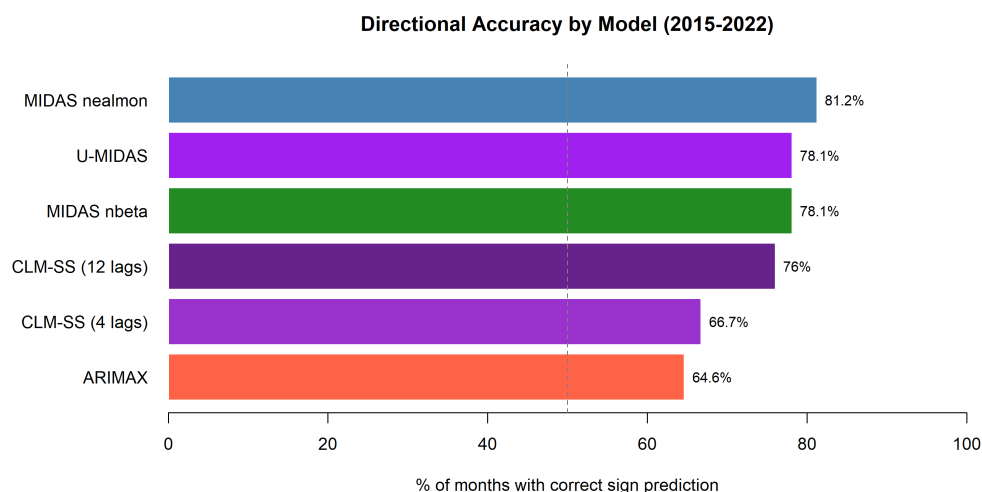


Figure 8: Directional accuracy by model, January 2015 to December 2022. MIDAS nealmon achieves the highest percentage of correct sign predictions in the original model-development period.

Figure 9 reports large-move precision and recall.

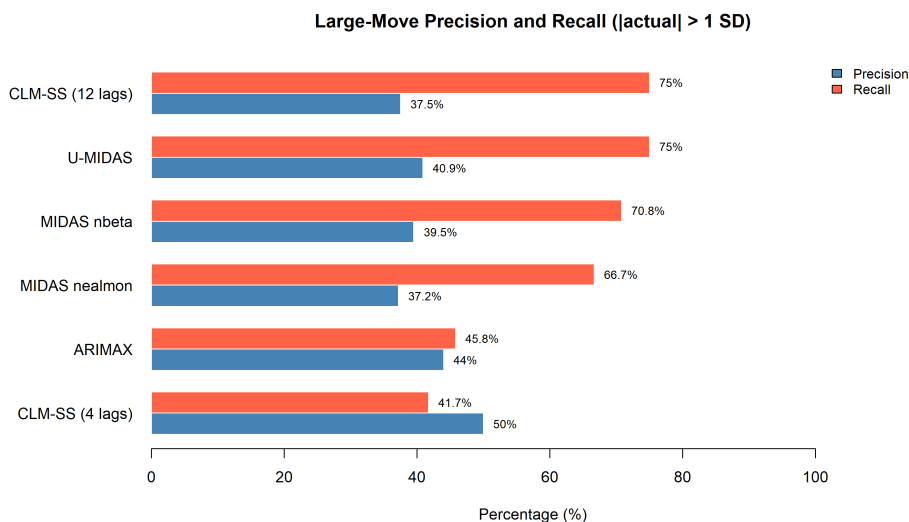


Figure 9: Large-move precision and recall for the main models. Precision is moderate across models, but recall is substantially stronger for the better weekly-timing specifications than for ARIMAX.

Figure 10 reports the Mincer–Zarnowitz beta coefficients.

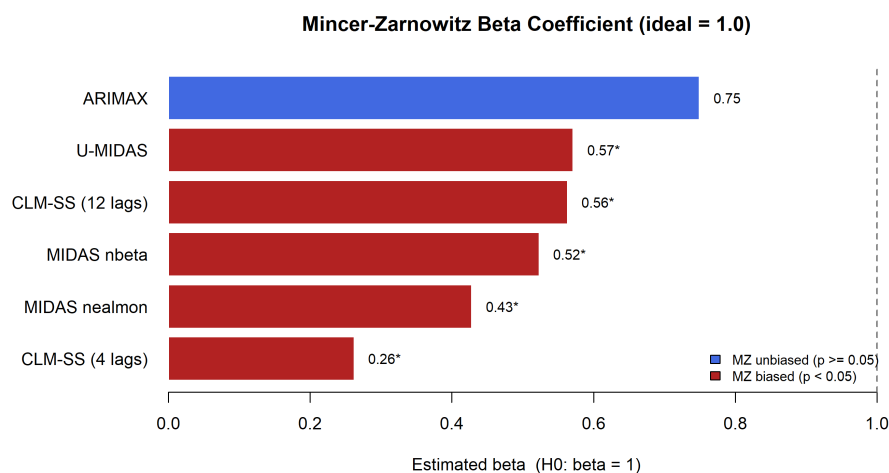


Figure 10: Mincer–Zarnowitz beta coefficients by model. ARIMAX appears closest to the ideal value of 1, but this must be interpreted together with predictive performance rather than in isolation.

C.2 Forecast-Horizon and Regime Robustness

Table 7 reports RMSE by horizon for ARIMAX, MIDAS nealmon, and MIDAS nbeta. MIDAS nbeta substantially outperforms at $h = 2$ (−31.8% versus ARIMAX), while rankings are less clear at $h = 1$ and $h = 3$.

Table 7: Forecast-Horizon RMSE for ARIMAX, MIDAS nealmon, and MIDAS nbeta

Horizon	ARIMAX	MIDAS nealmon	vs ARIMAX	MIDAS nbeta	vs ARIMAX
$h = 1$	0.02758	0.03453	+25.2%	0.03307	+19.9%
$h = 2$	0.03552	0.03515	−1.0%	0.02422	−31.8%
$h = 3$	0.02971	0.03306	+11.3%	0.03290	+10.7%

Tables 8 and 9 report regime-specific RMSE and MIDAS improvement over ARIMAX across the three periods. Mixed-frequency models are not uniformly best in every regime: MIDAS specifications are notably weaker during 2020 COVID, while XGBoost is the strongest model in that period.

Table 8: Regime-Specific Out-of-Sample RMSE (lower is better)

Model	2015–2019 pre-COVID	2020 COVID	2021–2022 recovery
ARIMAX	0.02412	0.02886	0.03425
LASSO-MIDAS	0.01842	0.03072	0.02704
MIDAS nealmon	0.01878	0.07591	0.02980
MIDAS nbeta	0.01940	0.05624	0.02952
XGBoost	0.02017	0.01456	0.02760
PCA-ARIMAX	0.02092	0.03484	0.03372

Table 9: MIDAS Improvement Over ARIMAX by Regime (% RMSE difference; negative = MIDAS better)

Regime	MIDAS nbeta	MIDAS nealmon
2015–2019 pre-COVID	−19.6%	−22.1%
2020 COVID	+94.9%	+163.1%
2021–2022 recovery/spike	−13.8%	−13.0%

C.3 Multicollinearity and Compression Diagnostics

Table 10 reports VIF diagnostics for all twelve weekly WTI lag regressors. All values are safely below the common concern threshold of 5, with a maximum of 1.79.

Table 10: VIF Diagnostics for Weekly WTI Lag Regressors (concern threshold = 5)

Lag	Month block	VIF
wti_m0_w1	Current month	1.47
wti_m0_w2	Current month	1.40
wti_m0_w3	Current month	1.46
wti_m0_w4	Current month	1.60
wti_m1_w1	Prior month	1.44
wti_m1_w2	Prior month	1.59
wti_m1_w3	Prior month	1.57
wti_m1_w4	Prior month	1.79
wti_m2_w1	Two months ago	1.23
wti_m2_w2	Two months ago	1.57
wti_m2_w3	Two months ago	1.49
wti_m2_w4	Two months ago	1.55

Figure 11 compares PCA-ARIMAX forecasts with MIDAS nbeta and ARIMAX.

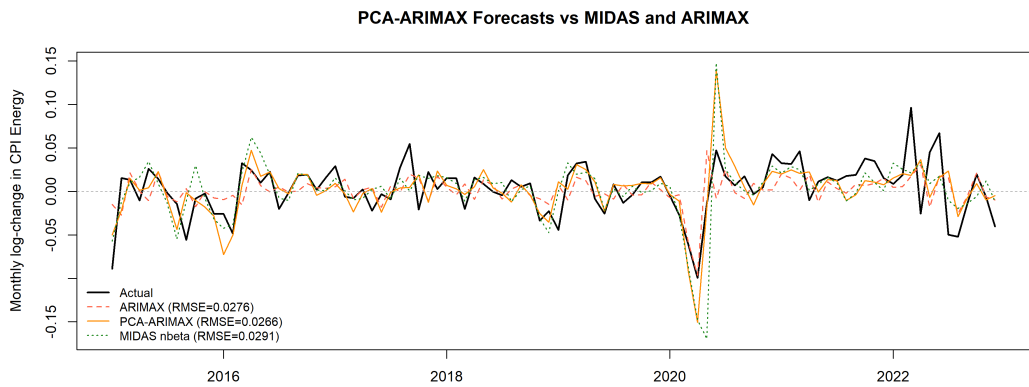


Figure 11: PCA-ARIMAX forecasts compared with MIDAS nbeta and ARIMAX. PCA compression improves slightly on the monthly benchmark, but does not recover the full gain from preserving weekly timing directly.

D Scenario-Based Future Forecasts

D.1 Short-Horizon Scenario Projections

The forward-looking exercise begins in July 2026 and reports 1-, 3-, and 6-month projections under alternative oil-price assumptions. These forecasts are not evaluated against known CPI outcomes and should therefore be interpreted as conditional scenario paths rather than as validated out-of-sample results.

Table 11 reports the terminal-month forecast CPI log-change for each scenario, horizon, and model. Values are the projected monthly log-change in US Consumer Energy CPI; positive values indicate an expected price increase and negative values an expected decrease.

Table 11: Short-Horizon Scenario-Based Future Forecasts

Horizon	Scenario	ARIMAX	MIDAS nbeta	U-MIDAS	XGBoost
1-month (2026-07)	Base: WTI flat	0.00371	0.00534	-0.00949	-0.01200
1-month (2026-07)	Recent mean	0.00911	0.01424	0.00062	-0.00377
1-month (2026-07)	Upside oil shock	0.00572	0.00865	-0.00573	-0.01247
1-month (2026-07)	Downside oil shock	0.00170	0.00203	-0.01324	-0.01426
3-month (2026-09)	Base: WTI flat	0.01003	0.00275	0.00226	-0.00153
3-month (2026-09)	Recent mean	0.01543	0.02453	0.02660	0.02371
3-month (2026-09)	Upside oil shock	0.01204	0.01085	0.01131	0.00744
3-month (2026-09)	Downside oil shock	0.00802	-0.00534	-0.00679	-0.00724
6-month (2026-12)	Base: WTI flat	0.00685	0.00275	0.00226	-0.00153
6-month (2026-12)	Recent mean	0.01225	0.02453	0.02660	0.02121
6-month (2026-12)	Upside oil shock	0.00886	0.01085	0.01131	0.00725
6-month (2026-12)	Downside oil shock	0.00484	-0.00534	-0.00679	-0.00718

E Code and Reproducibility

The complete R codebase for this capstone is available at https://github.com/stefhooy/midas_capstone. The repository contains the scripts, folder structure, and workflow used to generate the tables and figures reported in the thesis. The main scripts are:

- `08_energy_data.R`: prepares the energy data and mixed-frequency design.
- `09_benchmarks.R`: estimates ARIMAX, MIDAS, and U-MIDAS benchmarks.
- `07_clm_ss.R`: estimates the CLM-SS models.
- `10_ml_benchmarks.R`: estimates XGBoost, Kernel U-MIDAS, and LSTM.
- `11_extended_models.R`: runs LASSO-MIDAS, PCA/VIF diagnostics, segmented regression, and directional metrics.
- `12_final_evaluation.R`: generates the original final AIC/BIC and forecast-performance tables.
- `13_update_recent_data.R`: refreshes the dataset through 2026 and builds the later holdout design.
- `14_future_forecast_3_scenarios.R`: evaluates the post-2022 external holdout and produces the main 1-, 3-, and 6-month future scenario forecasts.
- `15_future_scenarios_12_months.R`: produces the 12-month appendix scenario forecasts and fan chart.

All figures and tables are saved in the `output/figures` and `output/tables` folders. Processed intermediate objects are saved in `data/processed`. A fixed random seed is used where relevant to make the results reproducible.

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